LUBS5050 Financial Derivatives
Sample Exam Question Paper
Financial Derivatives Sample Exam

Time allowed: 2 hours
Total marks: 100 marks
Instructions:

- Answer **ALL** questions in Section A.
- Answer **ONE** question from Section B.
- Answer **ONE** question from Section C.
- Show all workings for numerical questions.
- Assume continuous compounding unless otherwise stated.

Section A: Multiple Choice Questions

Total: 30 marks. Answer ALL questions. Each question is worth 2 marks.

Q1. A trader enters into a **short forward contract** to sell 200,000 euros at a forward exchange rate of $1.0800 per euro. At maturity, the spot exchange rate is $1.0950 per euro. What is the trader's gain or loss?

A. $3,000 gain
B. $3,000 loss
C. $15,000 gain
D. $15,000 loss
E. None of the above

Q2. Which statement best describes the difference between a forward contract and a call option?

A. A forward gives the right to buy; a call gives the obligation to buy.
B. A forward gives the obligation to buy or sell; a call gives the right to buy.
C. A forward is always exchange traded; a call is always OTC.
D. A forward always requires an upfront premium; a call does not.
E. None of the above.

Q3. Which of the following statements is correct?

A. Futures contracts are tailor-made agreements traded in OTC markets.

B. Forward contracts are standardized contracts traded on organised exchanges.

C. Futures contracts are marked to market daily and normally require margin accounts.

D. Forward contracts are always less risky than futures contracts because they do not require margin.

E. None of the above.

Q4. A trader enters into a long futures contract on 50,000 units of an asset at a futures price of \$12.40. At the end of the day, the futures price is \$12.10. What is the daily gain or loss?

A. \$15,000 gain

B. \$15,000 loss

C. \$30,000 gain

D. \$30,000 loss

E. None of the above.

Q5. A bank quotes an interest rate of 9% per annum with quarterly compounding. The equivalent continuously compounded rate is closest to:

A. 8.80%

B. 8.90%

C. 9.10%

D. 9.31%

E. None of the above.

Q6. A one-year long forward contract on a non-dividend-paying stock is entered into when the stock price is \$75 and the risk-free rate is 6% per annum with continuous compounding. Six months later, the stock price is \$78 and the risk-free rate is still 6% per annum. What are (i) the forward price when the contract is entered into and (ii) the value of the long forward contract after six months?

A. (i) \$77.28 and (ii) \$0.72

B. (i) \$79.64 and (ii) \$0.72

C. (i) \$77.28 and (ii) \$2.28

D. (i) \$79.64 and (ii) \$0.72

E. None of the above.

Q7. A stock index is currently 6,500. The risk-free rate is 4% per annum and the dividend yield is 2% per annum. What is the one-year futures price on the index?

A. 6,371

B. 6,500

C. 6,631

D. 6,765

E. None of the above.

Q8. A one-year forward contract on a non-dividend-paying stock was entered into six months ago. The delivery price is \$52. The current stock price is \$55 and the risk-free rate is 5% per annum. What is the current value of the long forward contract?

A. \$1.70

B. \$2.28

C. \$4.30

D. \$7.00

E. None of the above.

Q9. A company will sell an asset in three months and hedges using a futures contract. The futures price when the hedge is initiated is F_1, the spot price when the asset is sold is S_2, and the futures price when the futures position is closed out is F_2. The effective selling price under a short hedge is:

A. $F_1 - S_2$

B. $S_2 - F_1 + F_2$

C. $S_2 + F_1 - F_2$

D. $F_2 + S_2 - F_1$

E. None of the above.

Q10. A stock index is currently 7,200. The risk-free rate is 5% per annum, the dividend yield is 3% per annum, and the futures contract has six months to maturity. The standard deviation of changes in the spot price is 0.90, the standard deviation of changes in the futures price is 0.96, and the correlation coefficient between the two changes is 0.80. What are (i) the fair futures price and (ii) the minimum variance hedge ratio?

A. (i) 7,272 and (ii) 0.75

B. (i) 7,272 and (ii) 0.85

C. (i) 7,128 and (ii) 0.75

D. (i) 7,128 and (ii) 0.85

E. None of the above.

Q11. A portfolio is worth £25 million and has a beta of 1.2. The stock index futures price is 5,000 and each futures contract is worth £10 times the index. How many contracts should be used to fully hedge the portfolio?

A. Short 500 contracts

B. Short 600 contracts

C. Long 500 contracts

D. Long 600 contracts

E. None of the above.

Q12. For a European call option on a non-dividend-paying stock, the lower bound is:

A. $c \geq \max(S_0 - Ke^{-rT}, 0)$

B. $c \geq \max(Ke^{-rT} - S_0, 0)$

C. $c \leq S_0$

D. $c \leq Ke^{-rT}$

E. None of the above.

Q13. A European call option costs \$3.50. The stock price is \$40, the strike price is \$42, the risk-free rate is 5% per annum, and the time to maturity is one year. Using put-call parity, the price of the corresponding European put option is closest to:

A. \$1.45

B. \$3.45

C. \$5.45

D. \$7.45

E. None of the above.

Q14. A trader constructs a straddle by buying a call and buying a put with the same strike price of \$80. The call premium is \$5 and the put premium is \$4. The lower and upper break-even stock prices are:

A. \$71 and \$89

B. \$75 and \$84

C. \$76 and \$85

D. \$80 and \$89

E. None of the above.

Q15. A stock price is currently \$50. Over each of the next two six-month periods, it can either increase by 10% or decrease by 10%. The risk-free interest rate is 8% per annum with continuous compounding. What is the value of a one-year European call option with a strike price of \$50?

A. \$3.82

B. \$4.28

C. \$4.82

D. \$5.28

E. None of the above.

Section B: Structured Essay Questions

Total: 35 marks. Answer ONE question from this section.

Question 1

(a) Explain the difference between a **long forward position** and a **short forward position**. In your answer, explain the obligation created by each position and the direction of price movement that benefits each trader. [**10 marks**]

(b) Critically discuss the key differences between **exchange-traded derivatives** and **over-the-counter derivatives**. Your answer should refer to standardisation, flexibility, liquidity, counterparty risk, clearing arrangements, and examples of derivatives normally traded in each market. [**15 marks**]

(c) Explain why futures contracts can be used both for **hedging** and for **speculation**. Use examples to distinguish the motives of hedgers and speculators. [**10 marks**]

Question 2

(a) Explain what is meant by **basis risk** when futures contracts are used for hedging. Why does basis risk arise? [**10 marks**]

(b) A company expects to purchase a commodity in the future. Explain how it could use futures contracts to hedge this risk. In your answer, explain whether it should take a long or short futures position and why the hedge may not be perfect. [**10 marks**]

(c) Discuss the main issues that arise when a company uses a **stack-and-roll hedging strategy**. In your answer, refer to rollover risk, funding risk, margin calls, and the possible mismatch between the hedge instrument and the underlying exposure. [**15 marks**]

Section C: Options, Swaps, and Pricing

Total: 35 marks. Answer ONE question from this section.

Question 3

(a) Critically discuss the **comparative advantage argument** as an explanation for the popularity of swaps. Your answer should explain how swaps can transform the nature of liabilities or assets and how the total gain from a swap may arise. [**15 marks**]

(b) Explain the six main factors that affect the price of a **European call option**. For each factor, state whether an increase in that factor tends to increase or decrease the value of the call option. [**12 marks**]

(c) Explain the **put-call parity** relationship for European options on a non-dividend-paying stock. Why is this relationship important in no-arbitrage pricing? [**8 marks**]

Question 4

(a) Explain the difference between a **straddle** and a **strangle**. In your answer, discuss how each strategy is constructed, the market expectation behind each strategy, and the main difference in cost and break-even behaviour. [**10 marks**]

(b) Critically discuss the difference between a **strip** and a **strap** option strategy. In your answer, explain the investor's view about volatility and direction in each case. [**10 marks**]

(c) Explain the **risk-neutral valuation approach** to pricing a European option using a one-step binomial tree. Your answer should explain the meaning of the risk-neutral probability and why the real-world probability of an up movement is not required. [**10 marks**]

(d) Explain what is meant by **implied volatility**. What useful economic information can implied volatility provide, and what are its limitations? [**5 marks**]